

operations on integer data. Speaking of which, the data on which operations are performed in the CPU are called operands and they are of two types, integers and floating point values. And within floating point data, there are classifications based on the accuracy such as Single-Precision Floating-Point and Double-Precision Floating-Point formats. The sub-units within the processor that perform integer operations and those that perform floating-point operations are different.

The processor performs operations on data and this data has to be stored somewhere. Sometimes the data is obtained from the user and sometimes the data is generated by an element in the processor. It's the job of the memory management unit to ensure that the data required by the processor is always available to it. Since there are different layers of storage in each system ranging from the slow hard drives that are farthest from the ALU to the superfast registers that are nearest to it, it's the job of the Memory Management Unit to move the data to the appropriate storage layer based on when it's going to be required. Take for example that you're watching a movie that's stored on your hard drive. The processor tells the memory management unit that it's going to be decoding the data of the movie and then send

JARGON BUSTER

Active Heat Sink

An active heat sink uses moving components to keep a processor or other parts of a machine cool. This can include fans, and liquid cooling systems.

ALU

The arithmetic logic unit is a circuit used to perform maths and logic operations, and is the basic building block of a CPU.

Architecture

Depending on how the word is used, it can refer to the instruction set, the microarchitecture, or the process of fabrication.

Asymmetrical

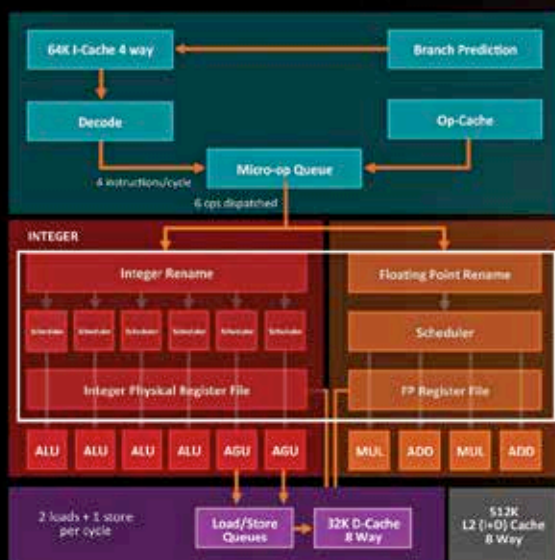
Asymmetric multiprocessing (AMP) systems are multi processor setups that reserve some functions for some processors, and do not treat them all equally.

it to the graphics processor so that it can be output on the screen. The data has to be moved from the hard drive to the RAM, then to the processor cache memory before finally reaching the registers. This movement of data is managed by the MMU.

Then comes the clock rate, which you're most familiar with because every CPU's clock rate is marketed the most. Remember seeing those numbers next to CPU models stating the GHz, MHz ratings? That's the clock rate. In the manner that we schedule our appointments and add structure to the way we operate by allocating a certain time period for each task, even processors do the same. The clock rate or clock speed or frequency are terms often used interchangeably to mean the same thing. Every step inside the processor requires a certain number of clock cycles for execution. This is where the whole concept of a processor clock = performance stems from. Though clock speed is an important factor, there are many more factors at play which have to be considered together to get a proper indication of the processor.

The Branch Predictor is a component within the processor that tries to predict the manner in which a sequence of instructions will progress before it's definitively known. This is

DESIGNING THE ENGINE: PERFORMANCE



QUANTUM LEAP IN CORE EXECUTION CAPABILITY

- ▲ Enhanced branch prediction to select the right instructions
- ▲ Micro-op cache for efficient op issue
- ▲ 1.75X instruction scheduler window*
- ▲ 1.5X issue width and execution resources*
- ▲ Result: instruction level parallelism designed for dramatic gains in single-threaded performance